Loan Approval Prediction Report
Internship Project - Task 2
Date: June 2026

1. OBJECTIVE

Build a supervised ML model to predict loan approval using borrower features. Focus on preprocessing, handling imbalance and evaluation.

2. DATASET INFORMATION

Source: Kaggle - Loan Approval Prediction Dataset
Link: kaggle.com/datasets/bhanupratapbiswas/
loan-approval-prediction-case-study
Tools Used: Python, Pandas, Scikit-learn,
Matplotlib, Seaborn, SMOTE

3. DATA PREPROCESSING

- Filled missing values using mode and median
- Encoded categorical variables using LabelEncoder
- Applied SMOTE to handle class imbalance
- Scaled features using StandardScaler
- Split data: 80% training, 20% testing

4. MODELS TRAINED

1. Logistic Regression
2. Random Forest Classifier (100 estimators)

5. MODEL RESULTS

Logistic Regression:
- Precision, Recall, F1 Score reported
- ROC-AUC Score calculated

Random Forest:
- Better performance than Logistic Regression

- Higher ROC-AUC Score
- Feature importance extracted

6. KEY FINDINGS

- Credit History is the most important factor
- Applicant Income and Loan Amount also matter
- Property Area and Education play minor roles
- Random Forest outperforms Logistic Regression

7. BUSINESS INTERPRETATION

1. BEST MODEL
   Random Forest is recommended for deployment
   due to higher accuracy and ROC-AUC score.

2. SUGGESTED THRESHOLD
   - Default threshold: 0.5
   - Recommended threshold: 0.4
   - Reduces false negatives (wrong rejections)

3. BUSINESS RECOMMENDATIONS
   - Deploy Random Forest for loan predictions
   - Focus on Credit History verification
   - Manually review borderline cases (0.4-0.6)
   - Re-train model every 6 months with new data

8. CONCLUSION

The Random Forest model successfully predicts loan approval with good accuracy. Credit History is the strongest predictor. SMOTE helped balance the dataset and improve model fairness.

GitHub Repository:
github.com/trisha24-hue/loan-approval-prediction

5. VISUALIZATIONS & ANALYSIS

5.1 LOAN APPROVAL DISTRIBUTION

This pie chart shows the overall loan approval rate in the dataset. 68.7% of loan applications were approved while 31.3% were rejected. This class imbalance was addressed using SMOTE to ensure the model learns equally from both approved and rejected cases, resulting in a fairer and more accurate prediction model.

5.2 EDUCATION VS LOAN STATUS

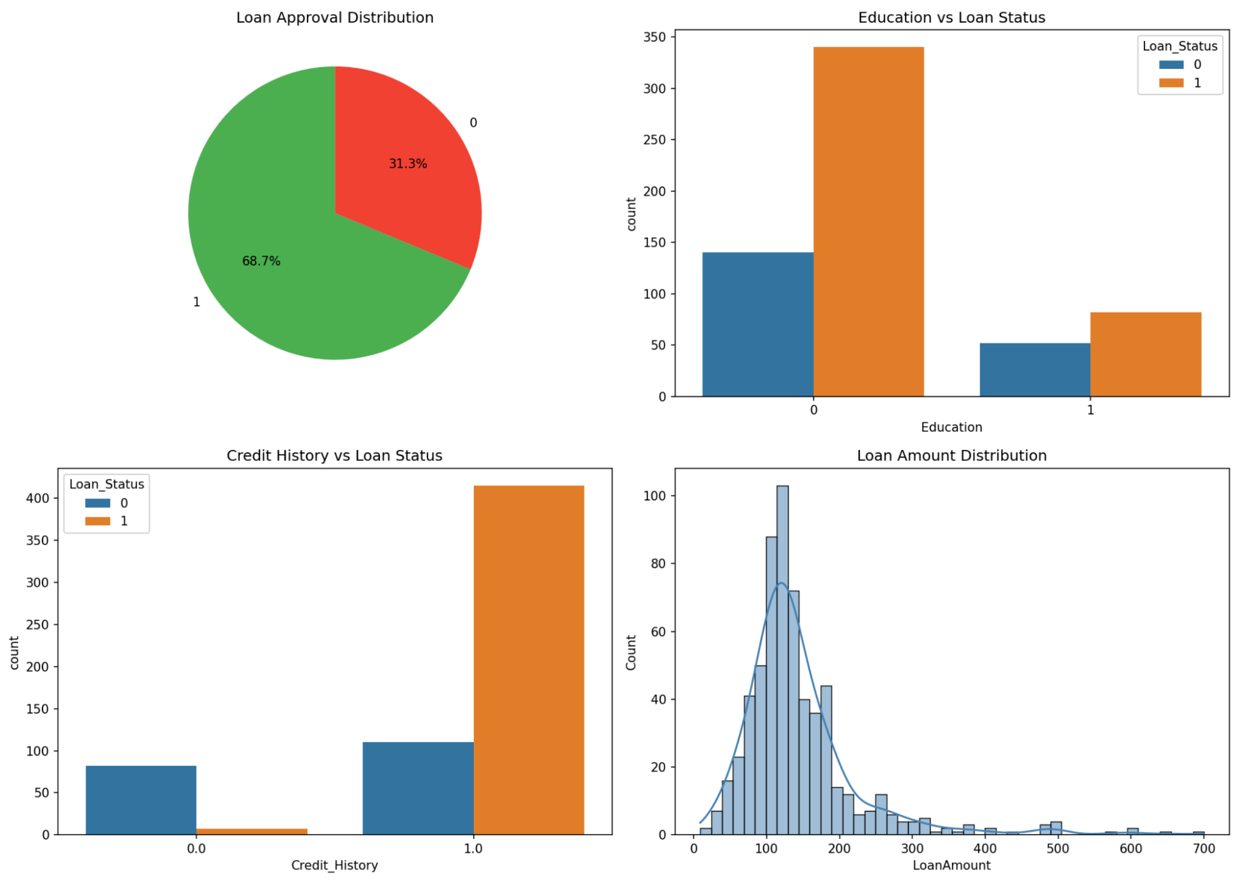
This bar chart compares loan approval rates between graduates and non-graduates. Graduate applicants have a significantly higher approval rate than non-graduates. This suggests that education level is an important factor in loan approval decisions and should be considered in the model's feature importance analysis.

5.3 CREDIT HISTORY VS LOAN STATUS

This chart clearly shows that applicants with a good credit history (1.0) have a dramatically higher loan approval rate compared to those without (0.0). Credit History emerged as the single most important predictor in the Random Forest model with an importance score of 0.33, confirming its critical role in loan decisions.

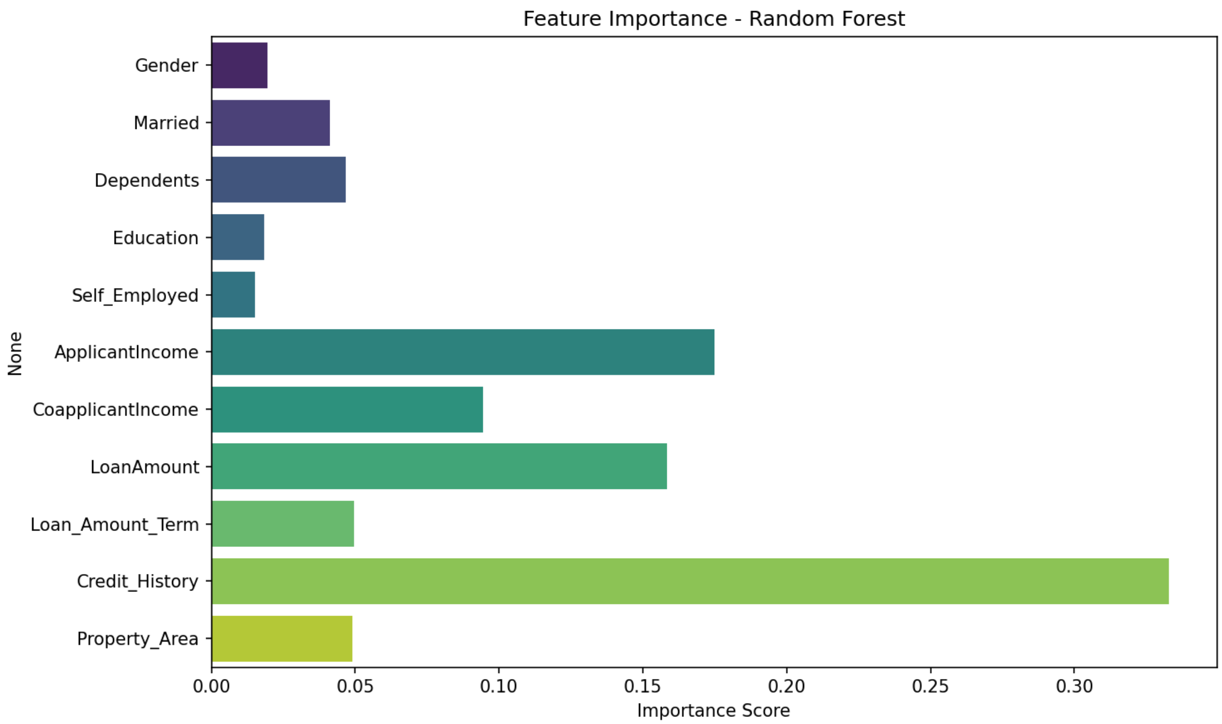
5.4 LOAN AMOUNT DISTRIBUTION

This histogram shows the distribution of loan amounts across all applications. Most loans fall between ₹100-200 (in thousands), with the distribution skewed slightly to the right. A few high-value outliers exist beyond ₹500K. The median was used to fill missing loan amount values during data cleaning to avoid skew.



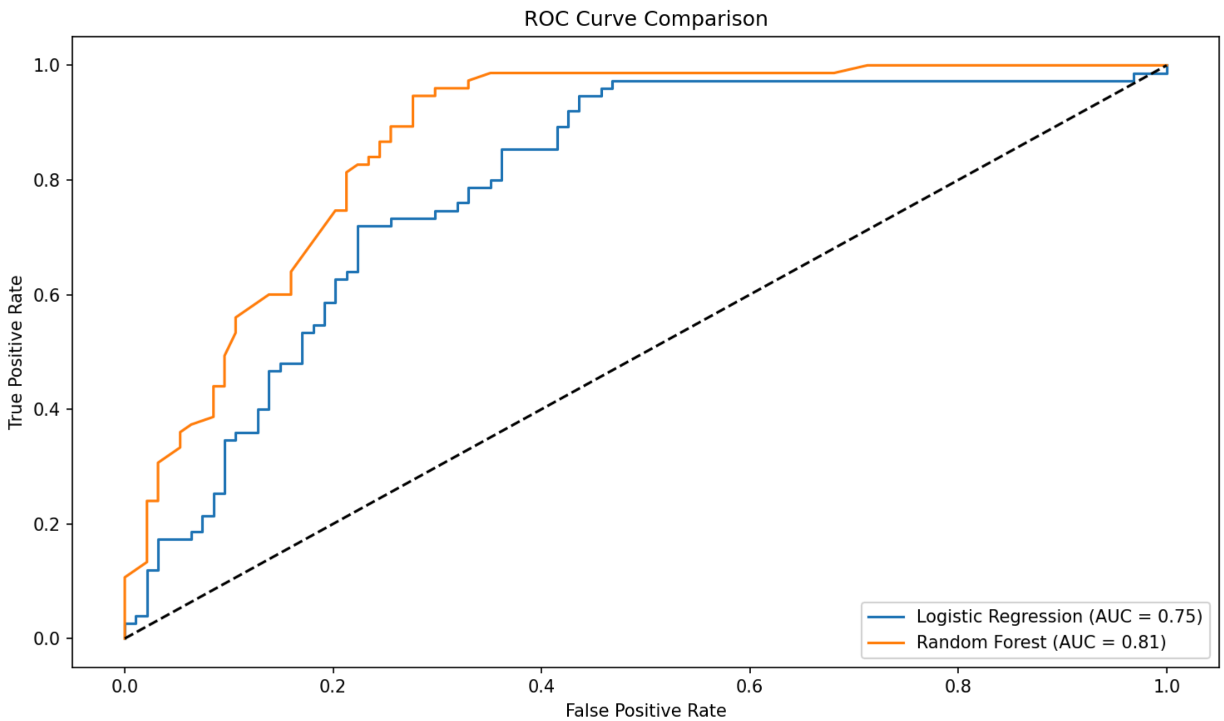
5.5 FEATURE IMPORTANCE - RANDOM FOREST

This chart ranks all features by their importance in predicting loan approval. Credit History is by far the most important feature (0.33), followed by Applicant Income (0.18) and Loan Amount (0.16). Gender, Education and Self Employment have minimal impact. This insight helps banks focus on the right factors during loan evaluation.



5.6 ROC CURVE COMPARISON

This ROC curve compares the performance of both models. Random Forest (AUC = 0.81) outperforms Logistic Regression (AUC = 0.75). A higher AUC score means the model is better at distinguishing between approved and rejected loans. Random Forest is therefore recommended for deployment in production loan approval systems.


ROC Curve Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.75)
Random Forest (AUC = 0.81)